

Short Communication

Effect of Si content on microstructure and mechanical properties of Al–Si–Mg alloys



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ABSTRACT

Microstructure and mechanical properties of as-cast and as-extruded Al–Si–Mg alloys with different Si content are investigated by tensile test, microstructure observation. High density of Si particles in the Al alloys can induce dynamic recrystallization during hot extrusion and it becomes more matured with an increase in the density of Si particles. The tensile strength of as-cast and as-extruded alloys can be improved with the increase of Si content and hot extrusion make the elongation of alloys increase dramatically. Considerable grain refining effect caused by recrystallization occurred during hot extrusion of S2 (equivalently commercial A356 alloy) and S3 (near eutectic alloy) alloys plays an important role in the improvement of elongation. A good combination of strength and elongation for the as-extruded S3 alloy indicates that near eutectic Al–Si alloys can be hot-extruded to produce aluminum profiles with high performance.

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1. Introduction

6XXX Al–Si–Mg aluminum alloys have been widely used as structural components, because of excellent formability, weldability and corrosion resistance [1]. However, low strength and rigidity make them prone to deformation failure in the process of application. Hypoeutectic and near eutectic Al–Si–Mg alloys, which have been usually used to produce castings, have higher strength and rigidity than 6xxx alloys. Nevertheless, the ductility of these alloys is decreased with increasing of Si content [2] and the brittle coarse Si particles usually make further deformation difficult. Na- or Sr-modification can change eutectic Si from coarse flakes into fine fibers, and after solutionized, the spheroidized eutectic Si particles are isolated in Al matrix [3,4]. Thus, a desired microstructure for further deformation could be obtained, implying that as-cast Al–Si–Mg alloys with high content of Si have good potential for wrought components.

In recent years, many researchers have focused their interests on the deformation behaviors of Al–Si base alloys. Generally, hot deformation can reduce defects which are formed during the solidification process and make Si particles distribute more uniformly [5], contributing to an increase in tensile strength. Some severe plastic deformation (SPD) methods can lead to a remarkable refinement of grains, such as RTMP (repeated thermomechanical processing) [6], HPT (high-pressure torsion) [7,8] and ECAP (equal-channel angular pressing) [9–12]. These methods were usually ap-

plied to obtain high ductility for Al–Si base alloys [6,7,10]. Nishida et al. [9] observed that dynamic recrystallization occurred during ECAP process of Al–7.0 mass% Si–0.35 mass% Mg alloy. A good combination of tensile strength and ductility could be achieved by using above-mentioned SPD methods. Fine-grained microstructure of Al matrix is thought to be the primary reason for the improvement of ductility [13,14]. However, researchers paid more attention on the improvement of mechanical properties while the evolution of microstructure and strengthening mechanisms are rarely mentioned. Gutierrez-Urrutia et al. [15] considered that Si particles may cause high density of dislocations which is responsible for the refinement of Al–7 wt.%Si alloy, but there are no further systematic studies to confirm this mechanism. Till now, the effect of Si content on microstructure evolution and mechanical properties of Al–Si base alloys has not been clearly established. In this paper, three Al–Si–Mg alloys with different amount of Si are prepared to investigate this effect by microstructure observation and tensile test.

2. Experimental procedures

2.1. Materials and processing

Three Al–Si–Mg alloys were prepared by controlled melting of commercial purity aluminum ingots, Al–12.5 wt.%Si master alloy and magnesium ingots (commercial impurity, 99.8 wt.%Mg) in a graphite crucible of 3 kg capacity using a melting/vacuum holding furnace. After processing, the melt was cast in a cylinder iron mold (the inner diameter of the ingots is 60 mm) into billets for further

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Table 1
Chemical composition (wt.%) of the experimental alloys.

Alloy	Si	Mg	Fe	Sr	Al
S1	0.58	0.51	0.15	–	Balance
S2	6.99	0.53	0.12	0.011	Balance
S3	12.30	0.47	0.15	0.020	Balance

hot extrusion and plate-like iron mold (with a cavity of $200 \times 100 \times 20 \text{ mm}^3$ for cast forming. These molds were preheated at 200°C for about 4 h. The composition of the experimental alloys measured by an ARL-3460 spectrum is listed in Table 1. S1 alloy (equivalently commercial 6063 aluminum alloy) has 0.58 wt.%Si and 0.51 wt.%Mg, not necessary to be modified with strontium, but S2 and S3 alloys were modified with 0.03 wt.% and 0.06 wt.% strontium respectively in the form of Al–10 wt.%Sr master alloy. After homogenized at 535°C for 6 h, the billets were immediately hot-extruded into $35 \text{ mm} \times 5 \text{ mm}$ sheets with an extrusion ratio of 16. In order to obtain the as-extruded microstructure, the sheets were quenched directly by water at the die exit.

2.2. Tensile test

The samples for tensile test were prepared from the as-cast plate-like castings and the as-extruded sheets according to GB/

T228-2002 [16] and the tensile test was carried out on a CMT4503 electronic universal testing machine at a constant cross-head speed of 2.0 mm min^{-1} . These samples were cut from the middle of the plate-like castings and the sheets along the extrusion direction.

2.3. Microstructure observation

Specimens for microstructural observation of as-cast, as-homogenized and as-extruded were cut respectively from the plate-like castings, cylinder billets and sheets along extrusion direction. They were polished using a standard metallographic procedure and then etched in 0.5% HF aqueous solution for 6 min. Optical microscope (OLYMPUS BX-60M) was used for the examination of microstructure and SEM (Philips XL30, Holland) studies were carried out on tensile fractured surfaces.

3. Results and discussion

3.1. Microstructure

It is well known that increasing of the Si content in cast Al–Si alloys increases the fraction of eutectic while decreases that of α -Al. Fig. 1a, c and e reveal this trend visually. In Fig. 1a, Mg_2Si and Fe-rich phases are observed at the grain boundaries due to

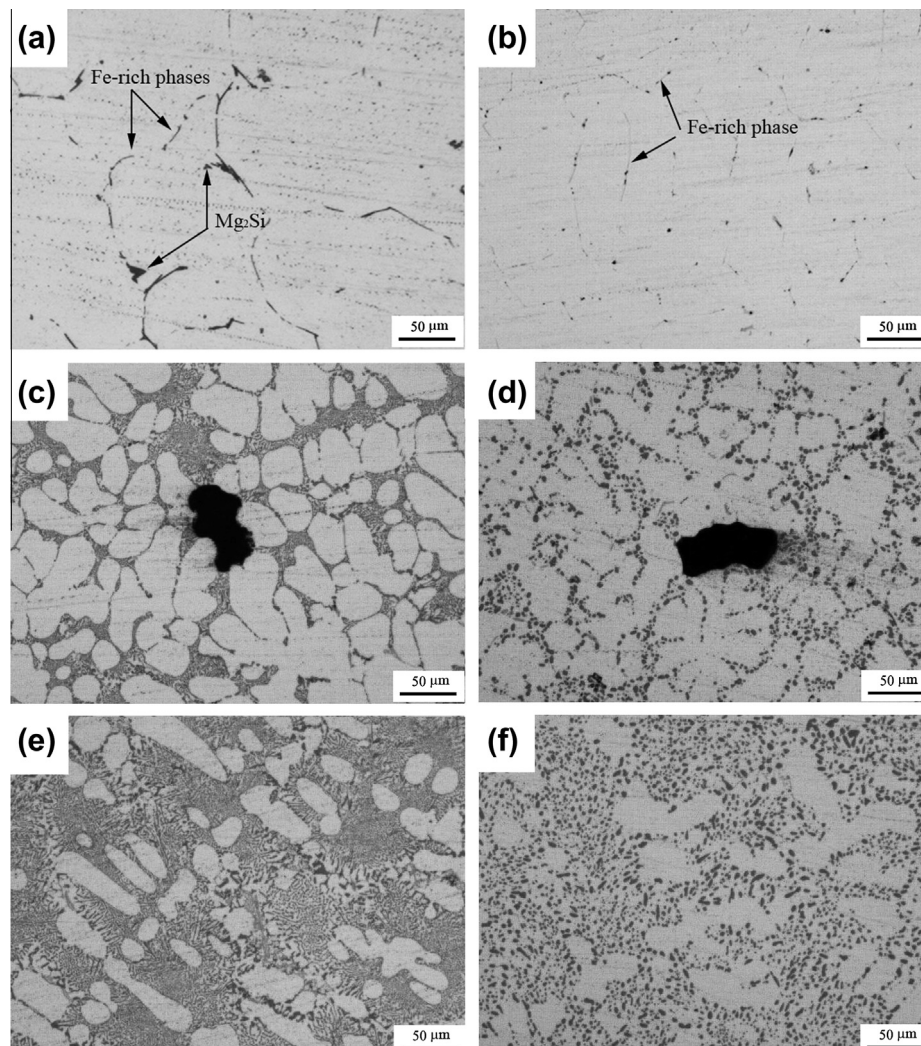


Fig. 1. Optical micrographs of three alloys: (a), (c) and (e) showing as-cast microstructures of S1, S2 and S3 alloys respectively; (b), (d) and (f) showing as-homogenized microstructures of S1, S2 and S3 alloys respectively.

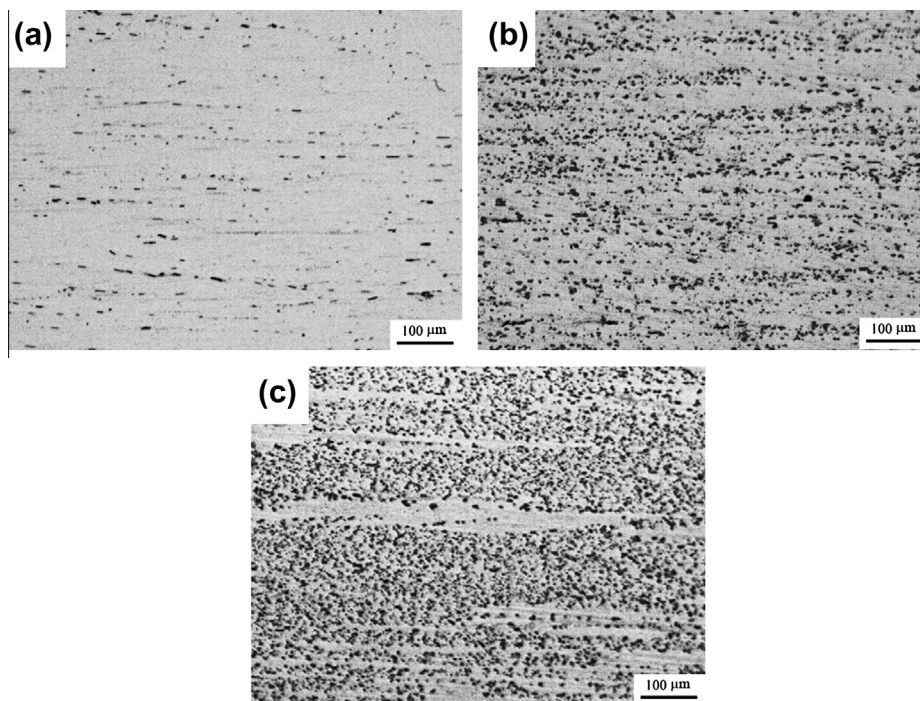


Fig. 2. The deformation traces of α -Al dendrites of as-extruded S1, S2 and S3 alloys.

non-equilibrium solidification in S1 alloy. Fig. 1c and e show typical as-cast micrographs of Al–Si cast alloys consisting of primary α -Al dendrites and interdendritic eutectic regions and the eutectic silicon are very fine and fibrous, suggesting a well modified microstructure. Fig. 1b, d and f represent the microstructures of three alloys after homogenized at 535 °C for 6 h before hot extrusion, Mg_2Si phase mainly disappears while the morphology of Fe-rich phases change into short rod-like considerably, as shown in Fig. 1b. For S2 and S3 alloys, the fibrous eutectic Si is well-spheroidized into fine particles which are isolated in Al matrix, as seen in Fig. 1d and f. However, it is worth noting that there are a lot of shrinkage- and gas-porosities in S2 alloy but almost no porosity is observed in S3 alloy. The large mushy zone of S2 alloy is thought to be responsible for the formation of porosity which has an important influence on mechanical properties.

Fig. 2 shows the micrographs of as-extruded S1, S2 and S3 alloys. Short rod-like Fe-rich phase distributes along the extrusion direction in S1 alloy, as shown in Fig. 2a. Primary dendrites in as-cast microstructures of S2 and S3 alloys are changed into bands owing to the plastic flow of the matrix along the extrusion direction. The bands of Si particle-rich and particle-poor correspond to original eutectic and primary α -Al dendrite areas respectively. In as-cast S2 alloy, the primary dendrites are dominant and the amount of it is much higher than that in as-cast S3 alloy. Thus, after hot extrusion, the amount of Si particle-poor bands (the deformation traces of α -Al dendrites) in S3 alloy is much less than that in S2 alloy and the width of them is also much smaller, as seen in Fig. 2. It is also seen that hot extrusion makes eutectic Si particles distribute more evenly than as-cast alloys.

Owing to the little amount of particles in S1 alloy, anodic film formation is a better choice to observe grain structure under polarizing microscope. For S1 alloy, band structure can be seen clearly and a small quantity of regular grains is formed along the band boundaries (Fig. 3), which suggests that recrystallization happens only near the grain boundaries. In S1 alloy, there are many survived Fe-rich particles at the grains boundaries (Fig. 2a). During hot extrusion, these particles will promote high deformation



Fig. 3. Polarized-light micrograph of as-extruded S1 alloy.

energy to accumulate in the matrix adjacent to the grain boundaries, resulting in recrystallization here. Because of the high content of Si particles in S2 and S3 alloys, it is very difficult to obtain a satisfied polarized-light micrograph. However, common optical micrographs can also reveal the recrystallized structure in as-extruded S2 and S3 alloys clearly. In S2 alloy, recrystallization has occurred during hot extrusion (as shown in Fig. 4a). Of more interest and importance, the grains in Si particle-rich bands are much finer than that in Si particle-poor bands, indicating that the recrystallization in particle-poor bands is not fully matured. As a result, the size of grains in S2 alloy is not uniform. Moreover, in S3 alloy, the grain structure (Fig. 4b) reveals a classical and mature recrystallization owing to the equally distributed Si particles. The above observation suggests that Si particles play a very important role in recrystallization. It is thought that during hot extrusion, the increased lattice distortion and accumulation of believed geometrically necessary dislocations around Si particles promote recrystallization occurring [17–19]. In S2 alloy, due to low local strain

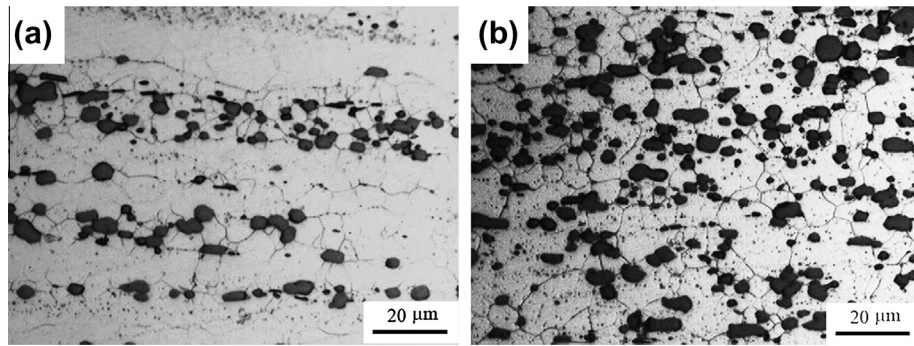


Fig. 4. Recrystallized grains in as-extruded alloys: (a) S2 alloy, (b) S3 alloy.

gradient in the matrix away from Si particles, the recrystallization is not fully matured. Nonetheless, in S3 alloy, Si particles with high density evenly distribute in Al matrix, thus the recrystallization is fully matured. Because the as-extruded sheets are water-quenched just out of the die-exit, the recrystallization occurred in S2 and S3 alloys is considered to be dynamic recrystallization [20]. Dynamic recrystallization of Al matrix has been found only in ECAP treated Al–Si alloys [21,22]. However, this phenomenon in hot extruded Al alloys has not been reported till now. From the above results, it can be concluded that the dynamic recrystallization can occur during hot extrusion of Al alloys containing a great amount of Si particles and it becomes more matured with an increase in the density of Si particles.

3.2. Mechanical properties

The tensile properties of the as-cast and as-extruded alloys are summarized in Fig. 5. As expected, with an increase in the Si content, the YS (yield strength) and UTS (ultimate tensile strength) of as-cast alloys are increased considerably. Compared with S1 alloy, YS and UTS of S2 alloy are improved by more than 30 MPa and 60 MPa respectively. However, with further increase in Si content, the YS and UTS of S3 alloy are only increased with about 20 MPa on the basis of S2 alloy. The similar effect has also been reported by Pedersen and Dwivedi et al. [2,23]. In addition, with increasing Si content, the elongation of as-cast alloys decreases as a whole. Wang et al. [24] and Kashyap et al. [25] have also found that an increase in Si content increases the strength, but at an expense of ductility. However, the elongation of as-cast S2 alloy is much less than that of S3 alloy. The porosity defects formed in S2 alloy (seen in Fig. 1c) should be responsible for it. The reason is that S2 alloy has a considerably wider mushy zone than S3 alloy, porosity defects are prone to form during solidification, which is detrimental to ductility [26].

The trend of as-extruded strength with ranging of Si content is similar to that of as-cast. Hot extrusion can eliminate porosity defects formed during solidification which is in favor of improving the mechanical properties, especially the ductility [27]. However, the remarkable changes in microstructures of the studied alloys after hot extrusion do not result in a considerable increase in tensile strength of as-extruded alloys, compared with as-cast alloys. In the as-cast alloys, due to non-equilibrium solidification, there is precipitation hardening effect of Mg_2Si phase to some extent. The homogenization process of 535 °C for 6 h before hot extrusion compels the solute of hardening phase to dissolve into aluminum matrix, and as a result, the precipitation hardening effect is removed. Thus, compared to as-cast alloys, hot extrusion does not lead to a considerable increase in strength even though the grain structure is refined remarkably. But it is worth noting that the elongation of three as-extruded alloys is improved significantly,

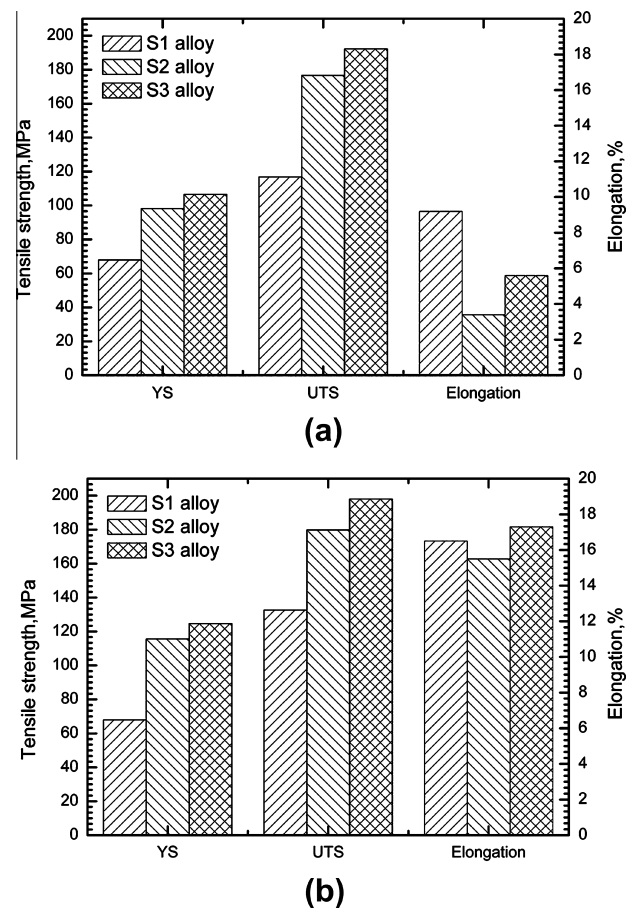


Fig. 5. Mechanical properties of the three alloys in two different conditions: (a) as-cast, (b) as-extruded.

from 9.2% (as-cast S1 alloy), 3.4% (as-cast S2 alloy) and 5.6% (as-cast S3 alloy) to 16.5% (as-extruded S1 alloy), 15.5% (as-extruded S2 alloy) and 17.3% (as-extruded S3 alloy) respectively. Contrary to the as-cast case, the elongation of as-extruded alloys does not decrease with the increase of Si content. The elongation of S3 alloy is definitely higher than S1 alloy. It is believed that considerable grain refining effect caused by recrystallization plays an important role in it. The related research [28] suggests that dispersoid/inclusion particles might lead to more inhomogeneous deformation and hence encouraged the faster accumulation of dislocations, as a result, additional plasticity was generated as a consequence of strain incompatibilities [18]. Nonuniformity of recrystallization and some existing defects in as-extruded S2 alloy (Figs. 2b and 4a) are responsible for lower elongation than S3 alloy.

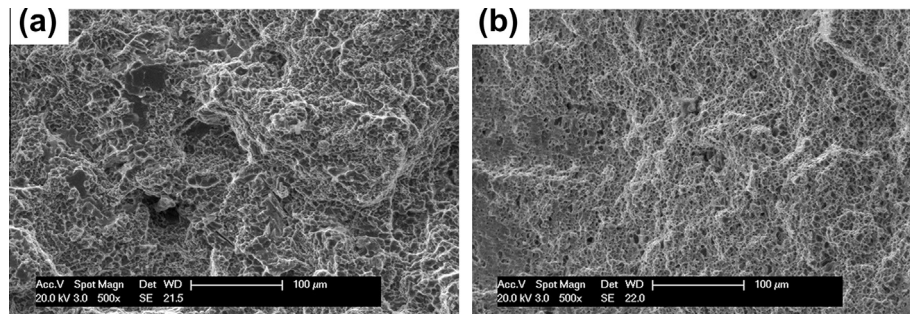


Fig. 6. Fractographs of S3 alloy: (a) as-cast, (b) as-extruded.

Fig. 6 shows the fractographs of S3 alloy under different conditions. There are some tear ridges in as-cast S3 alloy (Fig. 6a), which are typical characteristics of transgranular fracture. A great number of small dimples observed in as-extruded S3 alloy (Fig. 6b) indicate a more ductile fracture mode. From results above, it is concluded that the tensile strength of as-cast and as-extruded alloys can be improved with the increase of Si content and hot extrusion make the elongation of alloys increase dramatically. Furthermore, it implies that the near eutectic Al–Si alloys can be hot-extruded to produce aluminum profiles with high performance.

4. Conclusions

The effect of Si on microstructure and mechanical properties of three alloys is investigated by microstructure observation and tensile test. The following conclusions can be made:

- (1) Dynamic recrystallization of Al matrix can take place during hot extrusion of Al alloys containing a great number of Si particles and it becomes more matured with an increase in the density of Si particles.
- (2) The tensile strength of as-cast and as-extruded alloys can be improved with the increase of Si content and hot extrusion makes the elongation of the three alloys increase dramatically. Considerable grain refining effect from recrystallization occurred during hot extrusion of S2 and S3 alloys plays an important role in the improvement of elongation.
- (3) A good combination of strength and elongation for as-extruded S3 alloy indicates that near eutectic Al–Si alloys can be hot-extruded to produce aluminum profiles with high performance.

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